Java Full Stack Course Syllabus

Description: Java Full Stack curriculum from Java fundamentals through React, Spring Boot, databases, security, testing, and deployment.

Module 1: Java Fundamentals

Module Description: Learn Java syntax, setup, core programming concepts, and object-oriented basics.

Topic 1: Introduction and Setup

Topic Description:

Lesson 1: What is Java?

Lesson Description: Understand Java, the JVM, JRE, and JDK.

Lesson 2: Java Features

Lesson Description: Explore portability, security, and object-oriented programming.

Lesson 3: Installing JDK and IDE

Lesson Description: Set up Java with IntelliJ IDEA or Eclipse.

Lesson 4: First Java Program

Lesson Description: Compile and run a simple Java application.

Topic 2: Java Programming Basics

Topic Description:

Lesson 1: Variables and Data Types

Lesson Description: Use primitive types, variables, and type conversion.

Lesson 2: Operators and Expressions

Lesson Description: Apply arithmetic, relational, and logical operators.

Lesson 3: Control Statements

Lesson Description: Use if/else, switch, and loops.

Lesson 4: Arrays and Strings

Lesson Description: Work with arrays and common String operations.

Module 2: Core Java and Object-Oriented Programming

Module Description: Build reusable Java applications using OOP, collections, and exception handling.

Topic 1: Object-Oriented Java

Topic Description:

Lesson 1: Classes and Objects

Lesson Description: Create classes, constructors, methods, and objects.

Lesson 2: Encapsulation and Inheritance

Lesson Description: Organize and reuse code with access modifiers and inheritance.

Lesson 3: Polymorphism and Abstraction

Lesson Description: Use overriding, interfaces, and abstract classes.

Lesson 4: Exception Handling

Lesson Description: Handle checked and unchecked exceptions safely.

Topic 2: Java Libraries and Data

Topic Description:

Lesson 1: Collections Framework

Lesson Description: Use List, Set, Map, and iterators.

Lesson 2: Generics and Streams

Lesson Description: Write type-safe collections and process data with streams.

Lesson 3: File Handling

Lesson Description: Read and write files using Java I/O and NIO.

Lesson 4: JDBC Basics

Lesson Description: Connect Java applications to relational databases.

Module 3: Frontend Development

Module Description: Create responsive user interfaces using HTML, CSS, JavaScript, and React.

Topic 1: Web Foundations

Topic Description:

Lesson 1: HTML5 Structure

Lesson Description: Build semantic pages, forms, and accessible markup.

Lesson 2: CSS3 and Responsive Design

Lesson Description: Style layouts using Flexbox, Grid, and media queries.

Lesson 3: JavaScript Fundamentals

Lesson Description: Use functions, arrays, objects, DOM, and events.

Lesson 4: Modern JavaScript

Lesson Description: Work with ES modules, promises, async/await, and fetch.

Topic 2: React Frontend

Topic Description:

Lesson 1: React Components and JSX

Lesson Description: Build reusable UI components.

Lesson 2: Props, State, and Hooks

Lesson Description: Manage state and effects in React.

Lesson 3: Routing and Forms

Lesson Description: Create multi-page experiences and validate user input.

Lesson 4: API Integration

Lesson Description: Consume backend REST APIs and handle loading and errors.

Module 4: Backend Development with Spring Boot

Module Description: Build secure REST APIs with Spring Boot, persistence, and authentication.

Topic 1: Spring Boot and REST APIs

Topic Description:

Lesson 1: Spring Framework Basics

Lesson Description: Understand dependency injection and inversion of control.

Lesson 2: Spring Boot Setup

Lesson Description: Create and configure a Spring Boot application.

Lesson 3: Controllers and REST Endpoints

Lesson Description: Implement GET, POST, PUT, and DELETE endpoints.

Lesson 4: Service Layer and Validation

Lesson Description: Structure business logic and validate requests.

Topic 2: Persistence and Security

Topic Description:

Lesson 1: SQL and Database Design

Lesson Description: Write queries, joins, and relational schemas.

Lesson 2: Spring Data JPA and Hibernate

Lesson Description: Map entities and implement repositories.

Lesson 3: Spring Security

Lesson Description: Protect endpoints and manage authorization.

Lesson 4: JWT Authentication

Lesson Description: Implement token-based login and secured API access.

Module 5: Full Stack Integration, Testing and Deployment

Module Description: Connect frontend and backend, test features, and deploy a complete Java full stack project.

Topic 1: Integration and Quality

Topic Description:

Lesson 1: React and Spring Boot Integration

Lesson Description: Connect UI workflows to REST endpoints and configure CORS.

Lesson 2: Git and Maven

Lesson Description: Use version control, dependency management, and builds.

Lesson 3: Unit and API Testing

Lesson Description: Test Java services with JUnit/Mockito and verify REST endpoints.

Lesson 4: Error Handling and Logging

Lesson Description: Handle failures and collect useful application logs.

Topic 2: Project and Deployment

Topic Description:

Lesson 1: Full Stack CRUD Project

Lesson Description: Build a complete application with React, Spring Boot, and SQL.

Lesson 2: Authentication and Roles

Lesson Description: Implement login and role-based access across the stack.

Lesson 3: Production Configuration

Lesson Description: Manage environment settings and secrets.

Lesson 4: Deployment Basics

Lesson Description: Build and deploy frontend, backend, and database services.